CA Lab Exam - Section 3MCA06 - Set D

Cinema Movie List

Your task

The web page and the Express server are ready. The database part is missing. Complete it using Mongoose.

- Edit only **database.js**. The comments in it tell you what to write.
- Do not change any other file.

Steps

1. Open this project folder in VS Code and open a terminal in it.
2. Install the packages:

```
npm install
npm install mongoose
```

3. Make sure MongoDB is running, then add the sample data (see below).
4. Complete **database.js**.
5. Start the server:

```
node index.js
```

6. Open **http://localhost:3000** in your browser.

After every change to database.js, stop the server with Ctrl + C and start it again.

Database details

Database: cinema

Collection: movies

Fields: title (text), genre (text), year (number), rating (number)

Sample data

Open mongosh, paste these commands and press Enter. Paste them only once.

```
use cinema
db.movies.insertMany([
  { title: "Orbit Seven", genre: "Sci-Fi", year: 2014, rating: 7.8 },
  { title: "The Last Letter", genre: "Drama", year: 2009, rating: 8.1 },
  { title: "Laugh Lane", genre: "Comedy", year: 2018, rating: 6.9 },
  { title: "Mars Garden", genre: "Sci-Fi", year: 2021, rating: 8.4 },
  { title: "Quiet Harbour", genre: "Drama", year: 2016, rating: 7.2 },
  { title: "Office Chaos", genre: "Comedy", year: 2011, rating: 6.5 },
  { title: "Time Loop", genre: "Sci-Fi", year: 2008, rating: 7.5 },
  { title: "River of Dreams", genre: "Drama", year: 2022, rating: 8.7 },
  { title: "Wedding Mix-Up", genre: "Comedy", year: 2019, rating: 7.0 },
  { title: "Silent Planet", genre: "Sci-Fi", year: 2017, rating: 8.0 },
  { title: "Old Friends", genre: "Drama", year: 2012, rating: 7.6 },
  { title: "Pizza Party", genre: "Comedy", year: 2023, rating: 6.2 }
])
```

Show this output to the examiner

1. Terminal. Run node index.js.

Expected: Connected to MongoDB, then Server running at http://localhost:3000

2. Show all. Click "Show All Movies".

Expected: 12 movies.

3. Search. Choose Drama and click Search.

Expected: 4 movies: The Last Letter, Quiet Harbour, River of Dreams, Old Friends.

4. Filter. Type 2018 in "Released in or after" and click Filter.

Expected: 5 movies: Laugh Lane, Mars Garden, River of Dreams, Wedding Mix-Up, Pizza Party.

5. Top 3 Rated Movies. Click "Top 3 Rated Movies".

Expected: River of Dreams (8.7), Mars Garden (8.4), The Last Letter (8.1), in this order.

6. Add Movie. Fill the form with Title: Exam Day, Genre: Comedy, Year: 2026, Rating: 7.4. Click "Add Movie".

Expected: The message "Movie saved" appears and the list shows 13 movies, including Exam Day.